Practice workbook · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Defining and recognising attacks, spotting phishing indicators in an email, and explaining how an attack works or is prevented.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-09

1. Question 2 · part c.i · 2 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

2 Danielle manages network security for a small business.

(a) Identify **one** action Danielle could take to protect the **network** from unauthorised access.

(1)

- ☒ **A** Give each user a separate account
- ☒ **B** Enforce a strong password policy
- ☒ **C** Make a backup plan
- ☒ **D** Encrypt sensitive files

(b) Danielle has received this email.

She is a customer of SafePayment.com but thinks this is a phishing attempt.

From: SafePayment.accounts@5afepayment.com
To: Danielle616
Subject: Account restricted (case SP-0011312-2021-06)

Dear Customer,

We have noticed some unusual activity on your account, so have stopped all payments.

Please use this link to log in and check your account.

Log in

Your normal log in will stoop working until you have done this.

Regards
SafePayment fraud prevention team.

(i) Danielle has circled the words **Dear Customer**.

Give **one** reason why the use of these words might make Danielle suspect that the email is a phishing attempt.

(1)

(ii) Draw circles around **two other** aspects of the email that might make Danielle suspect it is a phishing attempt.

(2)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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P 6 9 2 9 7 A 0 6 2 0

1. Question 2 · part c.i · 2 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

DO NOT WRITE IN THIS AREA

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(c) Some employees use company laptops in public places.

Danielle is concerned that shoulder surfing could pose a risk to security.

(i) Describe what is meant by **shoulder surfing**.

(2)

(ii) Explain **one** way to prevent shoulder surfing.

(2)

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(d) Danielle wants to protect the network from cyberattacks.

(i) Identify the purpose of penetration testing.

(1)

- ☒ **A** To decompile executable code
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P 6 9 2 9 7 A 0 7 2 0

7

Turn over ▶

1. Question 2 · part c.i · 2 marks · Working and answer

Working and answer

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2. Question 2 · part b.i · 1 mark

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Further practice · Attempt part b.i only.

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2. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

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3. Question 2 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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2021 NOV · Paper 01

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Working and answer

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4. Question 2 · part c.ii · 2 marks

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4. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

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5. Question 2 · part d.ii · 2 marks

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5. Question 2 · part d.ii · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part d.ii only.

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P 6 9 2 9 7 A 0 7 2 0

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Turn over ▶

5. Question 2 · part d.ii · 2 marks

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Further practice · Attempt part d.ii only.

(ii) Danielle is sent a software patch for the network operating system but is too busy to install it straight away.

Explain why this delay could pose a threat to the security of the network.

(2)

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(Total for Question 2 = 11 marks)

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Working and answer

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6. Question 5 · part c · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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6. Question 5 · part c · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c only.

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(c) The network is at risk of an eavesdropping attack.

Identify the description of eavesdropping.

(1)

- ☒ **A** Tricking people into giving information by sending emails pretending to be from someone in authority
- ☒ **B** Spying on someone using a computer
- ☒ **C** Intercepting information as it is transmitted over a network
- ☒ **D** Redirecting a user from a genuine website to a fake one

(d) Doctors use laptops when they visit patients in their homes.

(i) The laptops have solid state drives.

Explain **one** reason why a solid state drive is better than a magnetic hard drive for the laptops.

(2)

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(ii) Describe how data is stored on a solid state drive.

(2)

(iii) The laptops have two types of memory.

Complete the table by adding **one** tick (✓) to match **each** description to the type of memory used.

(2)

Description	RAM	ROM
Stores the boot up sequence		
The contents are lost when the laptop is shut down		

(Total for Question 5 = 12 marks)

6. Question 5 · part c · 1 mark · Working and answer

Working and answer

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7. Question 1 · part b.iii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.iii only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

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7. Question 1 · part b.iii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.iii only.

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(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

(2)

1 .....

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2 .....

.....

(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

(1)

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.....

(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

(1)

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

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7. Question 1 · part b.iii · 1 mark · Working and answer

Working and answer

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